

STUDY GUIDE

Introductory Molecular Biology

BCH 216

A complete, exam-ready revision guide

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1 Nucleic Acid Chemistry

1.1 Introduction

Nucleic acids are macromolecules present in all living cells; when combined with proteins they form **nucleoproteins**. Their central role is the storage and transfer of genetic information. They are polymers of nucleotides joined by **phosphodiester bonds**, and are therefore described as **polynucleotides**.

Nucleotides are not only the monomeric building blocks of nucleic acids; they also underpin much of cellular biochemistry. ATP (adenosine triphosphate) is the universal currency of energy; nucleotides are components of key coenzymes such as NAD^+ and FAD; and derivatives such as cyclic AMP (cAMP) act as metabolic regulators.

1.2 Types of Nucleic Acid

There are two principal types: **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**. DNA is predominantly located in the nucleus, with small amounts in the mitochondria; about 90% of RNA is found in the cytoplasm and 10% in the nucleolus.

DNA stores and transmits genetic information and carries the instructions for growth, development, reproduction and cell function across all organisms, from single-celled bacteria to multicellular mammals. RNA transfers that information from DNA to the protein-synthesising machinery. Some viruses use RNA rather than DNA as their genetic material, but are not strictly considered alive because they cannot reproduce without a host.

1.3 Nucleosides and Nucleotides

A **nucleoside** is composed of a nitrogenous base (a purine or pyrimidine) plus a pentose sugar (ribose or 2-deoxy-D-ribose). A **nucleotide** adds one, two or three phosphate groups to that nucleoside.

Bases at a glance

Purines (double ring): Adenine, Guanine. **Pyrimidines** (single ring): Cytosine, Thymine, Uracil. Cytosine occurs in both DNA and RNA; **Thymine is DNA-only**; **Uracil is RNA-only**.

The base is joined to C1 of the pentose by an **N-glycosidic bond** - N9 of a purine, or N1 of a pyrimidine.

The tables below name the nucleosides and nucleotides formed from each base.

Ribonucleosides (base + D-ribose)

Base	Pentose sugar	Nucleoside
Adenine	Ribose	Adenosine
Guanine	Ribose	Guanosine
Uracil	Ribose	Uridine
Cytosine	Ribose	Cytidine
Hypoxanthine	Ribose	Inosine

Base	Pentose sugar	Nucleoside
Xanthine	Ribose	Xanthosine

Deoxyribonucleosides (base + 2-deoxy-D-ribose)

Base	Pentose sugar	Nucleoside
Adenine	Deoxyribose	Deoxyadenosine
Guanine	Deoxyribose	Deoxyguanosine
Cytosine	Deoxyribose	Deoxycytidine
Thymine	Deoxyribose	Deoxythymidine

Deoxyribonucleotides (base + deoxyribose + phosphate)

Base	Sugar	Phosphate	Nucleotide
Adenine	Deoxyribose	Pi	Deoxyadenosine monophosphate (dAMP)
Guanine	Deoxyribose	Pi	Deoxyguanosine monophosphate (dGMP)
Cytosine	Deoxyribose	Pi	Deoxycytidine monophosphate (dCMP)
Thymine	Deoxyribose	Pi	Deoxythymidine monophosphate (dTMP)

1.4 Biologically Important Nucleotides

ATP is the main biological source of energy, required in pathways such as fatty-acid synthesis, glycolysis, cholesterol synthesis, protein synthesis and gluconeogenesis, and in functions such as muscle contraction and nerve-impulse transmission. **AMP** is a component of coenzymes including NAD^+ , NADP^+ , FAD and coenzyme A. **cAMP** (cyclic adenosine 3',5'-monophosphate), formed from ATP by adenylate cyclase, is a second messenger for many hormones such as epinephrine and glucagon.

UDP participates in glycogenesis; UDP-glucose and UDP-galactose function in galactose metabolism and in the synthesis of lactose and cerebrosides, while UDP-glucuronic acid is needed for detoxification and for mucopolysaccharide synthesis (e.g. heparin, hyaluronic acid). **cGMP**, formed from GTP by guanylyl cyclase, is an intracellular second messenger that can act antagonistically to cAMP and is involved in smooth-muscle relaxation and vasodilation.

1.5 DNA Structure and the Watson-Crick Model

DNA is a polymer of deoxyribonucleotides linked covalently by **3',5'-phosphodiester bonds**: the 3'-hydroxyl of one sugar is joined to the 5'-hydroxyl of the adjacent sugar. In eukaryotes DNA resides in the nucleus and in organelles such as mitochondria and chloroplasts, organised into long linear chromosomes; in prokaryotes it lies, unenclosed by a membrane, in a region called the nucleoid.

In 1953 James Watson and Francis Crick deduced the three-dimensional structure of DNA. Its key features are summarised below.

Feature	Description
Right-handed double helix	Two polydeoxyribonucleotide chains twist around one another like a spiral staircase; the sugar-phosphate groups form the handrails and the bases the steps.
Base-pairing rule	Adenine pairs with Thymine (2 H-bonds); Guanine pairs with Cytosine (3 H-bonds). Strands are complementary.
Hydrogen bonding	Strands are held mainly by H-bonds between bases; the G-C bond is stronger than the A-T bond.
Antiparallel strands	One strand runs 5' to 3', the other 3' to 5'.

Chargaff's Rule

In the DNA of all species, the quantity of purines equals that of pyrimidines: $A + G = T + C$.
 Content of Adenine equals Thymine ($A = T$) and Guanine equals Cytosine ($G = C$).
 The ratio of purine to pyrimidine bases is therefore always one.

Denaturation (melting): double-stranded DNA can be separated by heat. The **melting temperature** is the temperature at which half the helical structure is denatured. On cooling, the separated strands re-associate in a process called **annealing**.

1.6 Ribonucleic Acid (RNA)

RNA is usually single-stranded. Each nucleotide carries one of four bases (A, U, G or C), a ribose sugar and a phosphate group. Thymine is replaced by **Uracil**: a DNA coding strand 5'-AATTGCGC-3' corresponds to the RNA sequence 5'-AAUUGCGC-3'.

Type of RNA	Role
Messenger RNA (mRNA)	Intermediate between the gene in DNA and the protein-synthesising machinery; carries the message to be translated into protein.
Ribosomal RNA (rRNA)	Major component of ribosomes; helps mRNA bind correctly. Some rRNAs act as enzymes (ribozymes).
Transfer RNA (tRNA)	Carriers that bring amino acids to the ribosome, ensuring the correct amino acid specified by the mRNA is added.
Regulatory RNA	RNA species that regulate gene expression.

2 Biochemistry of Heredity and Genetics

2.1 Introduction

Genetics is the branch of biology that studies how traits pass from parents to offspring. This passing of traits is called **heredity**, so genetics is the study of heredity. The discipline is built around DNA, which holds all the genetic information for an organism and provides cells with the instructions they need to grow, survive and reproduce.

A **gene** is a particular section of a DNA molecule that directs a cell to perform one specific task. During reproduction, DNA is replicated and passed from parent to offspring; this inheritance influences the appearance and behaviour of the offspring, although the environment can also influence how genes are expressed.

Why study genetics?

Diagnose, treat, prevent and cure illness - genes are instructions for the proteins the body needs. Understand human history, and support forensic and legal applications (e.g. paternity, court cases).

Develop targeted therapies and explore genetic enhancement.

2.2 Genes and Alleles

In 1865 **Gregor Mendel** first described the elements of heredity. From his study of the pea plant he concluded that specific traits pass unchanged from one generation to the next. Genes vary in size from a few hundred to more than two million DNA bases; most are the same in all people, but a small number differ between individuals. Each human nucleated somatic cell contains about 30,000 genes in the nucleus, plus some non-nuclear genes in the mitochondria.

Alternative forms of a gene are called **alleles**. For each gene an individual inherits one allele from each parent, and so has two alleles at each autosomal locus and at each X-chromosome locus in females. Males have only one X chromosome and are therefore **homozygous** (hemizygous) for all X-linked genes.

Term	Meaning
Dominant allele	Shows its effect even with a single copy; masks the recessive allele. Written as a capital letter (e.g. B).
Recessive allele	Shows its effect only when two copies are present. Written as a lower-case letter (e.g. b).
Homozygous	Two identical alleles at a locus (e.g. BB or bb).
Heterozygous	Two different alleles at a locus (e.g. Bb).
Genotype	The genetic constitution - what the genes 'say'.
Phenotype	The physical expression of the genotype - what you see.

Worked example - eye colour

BB = Brown eyes (homozygous dominant)

Bb = Brown eyes (heterozygous - still brown, because B is dominant)

bb = Blue eyes (homozygous recessive)

2.3 Chromosomes

Within the nucleus, DNA is packed into thread-like structures called **chromosomes**, each made of DNA coiled many times around proteins called **histones**. Chromosomes become visible under the microscope only during cell division, when the DNA is most tightly packed. Each chromosome has a constriction point, the **centromere**, dividing it into a short arm (**p arm**) and a long arm (**q arm**).

There are two types of chromosome. **Autosomes** carry genes for most body traits - humans have 22 pairs. **Sex chromosomes** determine sex: XX in females and XY in males. In total humans have 23 pairs (46 chromosomes): 22 pairs of autosomes and one pair of sex chromosomes.

Ploidy

Haploid (n): a single set of chromosomes - in humans, 23. Produced by meiosis (e.g. gametes).

Diploid (2n): two sets - in humans, 46. The normal complement of somatic cells.

2.4 The Process of Heredity

Heredity is made possible by chromosomes, the vehicles of genetic inheritance described by the **chromosomal theory of inheritance**. In sexually reproducing species the process involves gametes (sperm and egg) and **meiosis**: a diploid cell with 46 chromosomes divides to produce four haploid gametes of 23 chromosomes each.

At **fertilisation** a sperm and egg fuse, combining their chromosomes into a diploid zygote with 46 chromosomes - half from each parent. Meiosis, including **crossing over** (the exchange of segments between homologous chromosomes), generates an enormous number of unique genetic combinations, ensuring that offspring are genetically distinct from their parents.

2.5 The Human Genome Project (HGP)

The HGP was one of the most significant research endeavours of the twentieth century. Its goals included identifying all human genes, building genetic-linkage and physical maps, determining the sequence of the ~3 billion base pairs of human DNA, storing the data in public databases, developing analysis tools, and transferring technologies to the private sector.

In April 2003 researchers announced a high-quality sequence of essentially the entire human genome, closing gaps in the 2001 working draft, locating many genes and freely publishing the data online. The project also sequenced the genomes of other organisms (brewer's yeast, the roundworm, the fly, and a working draft of the mouse) to enable comparative study. An **Ethical, Legal and Social Implications (ELSI)** programme examined fairness, privacy, psychological impact, reproductive and clinical issues, and more.

Genome facts uncovered

The genome contains about 3.2 billion A, C, G and T bases.

An average gene has ~3,000 bases; the largest known gene, **dystrophin**, has ~2.4 million bases.

Chromosome 1 holds the most genes (~3,168); the Y chromosome the fewest (~344).

2.6 Blood Groups - The ABO and Rh Systems

A blood group classifies blood by the presence or absence of inherited **antigens** on the surface of red blood cells. The two most important systems are **ABO** and **Rh**, which together create the eight main blood types. Knowing one's blood type is crucial for safe transfusion and for managing pregnancy.

ABO type	Antigens on RBC	Antibodies in plasma	Possible genotype
A	A	anti-B	AA or AO
B	B	anti-A	BB or BO
AB	A and B	none	AB
O	none	anti-A and anti-B	OO

The A and B alleles are **dominant** over the recessive O allele, and A and B are **codominant** (an AB genotype expresses both). For the Rh system, the Rh-positive gene is dominant, so a person must inherit two recessive genes to be Rh-negative.

The Rh (Rhesus) system

The Rh factor is an inherited protein on the red-cell surface. Rh-positive individuals carry it; Rh-negative individuals lack it. Combining ABO and Rh gives the eight common types (A+, A-, B+, B-, AB+, AB-, O+, O-).

Feature	Rh positive	Rh negative
Protein on red cells	Present	Absent
Occurrence	More common (~85%)	Less common (~15%)
Donation compatibility	Can receive Rh+ or Rh-	Can receive only Rh-
Pregnancy concern	Usually none	Possible Rh incompatibility

Transfusion compatibility

Universal red-cell donor: O-negative (lacks A, B and Rh antigens). **Universal red-cell recipient:** AB-positive (no A, B or Rh antibodies). **Universal plasma donor:** AB plasma (lacks anti-A and anti-B). If an Rh-negative person receives Rh-positive blood, the immune system may treat the Rh protein as foreign and produce antibodies, causing a dangerous transfusion reaction.

Significance in pregnancy

Rh incompatibility is most critical when an Rh-negative woman carries an Rh-positive fetus. Maternal and fetal blood do not normally mix, but a small mixing during delivery, miscarriage or trauma can prompt the mother to make anti-Rh antibodies. The first Rh-positive baby is

usually unaffected, but in a subsequent Rh-positive pregnancy the pre-formed antibodies can cross the placenta and attack fetal red cells, causing **haemolytic disease of the newborn (HDN)** - severe anaemia, jaundice, brain damage, or even stillbirth.

Prevention

Rh-negative women can be given **Rh immunoglobulin (RhIg / RhoGAM)**, typically around 28 weeks of pregnancy and again shortly after delivering an Rh-positive baby, preventing the mother from producing the harmful antibodies and protecting future pregnancies.

3 DNA Replication, Transcription and Protein Synthesis

3.1 DNA Structure (Recap)

DNA usually exists as a double-stranded double helix. Each strand is a chain of four nucleotides carrying the bases Adenine, Cytosine, Guanine and Thymine (A, C, G, T). **Adenine pairs with Thymine** via two hydrogen bonds; **Cytosine pairs with Guanine** via three. A purine must always pair with a pyrimidine - a purine cannot pair with another purine because the strands would crowd together, and a pyrimidine-pyrimidine pair would leave an unbound hydrogen, destabilising the DNA. Each strand has directionality, named from the 5' end to the 3' end.

3.2 Key Enzymes of DNA Replication

Enzyme	Function in DNA replication
DNA helicase	Unwinds the DNA double helix at the replication fork.
DNA primase	Provides a starting RNA/DNA primer for DNA polymerase to begin synthesis.
DNA polymerase	Builds the new strand by adding nucleotides 5' to 3'; also proofreads and corrects errors.
DNA clamp	Prevents DNA polymerase III from dissociating from the parent strand.
Single-strand binding proteins (SSB)	Bind ssDNA and prevent re-annealing, maintaining strand separation.
Topoisomerase	Relaxes DNA from its supercoiled state.
DNA gyrase	Relieves the strain of unwinding caused by helicase.
DNA ligase	Re-anneals the strand and joins Okazaki fragments of the lagging strand.
Telomerase	Lengthens telomeric DNA by adding repetitive sequences to chromosome ends.

3.3 The Replication Process

DNA replication copies DNA and is the basis of biological inheritance; without it, each daughter cell would receive only half a genome. One double-stranded molecule produces two identical copies, with each parent strand serving as a template for a complementary new strand - a mechanism termed **semi-conservative replication**. Replication begins at specific sites called **origins**; unwinding at the origin forms a **replication fork**, where DNA polymerase adds nucleotides matched to the template.

DNA polymerase reads the template 3' to 5' and so synthesises the new strand 5' to 3'. Because the two template strands are antiparallel, the two new strands are made differently:

Strand	How it is synthesised
Leading strand	Synthesised continuously, in the same direction as the advancing replication fork.

Strand	How it is synthesised
Lagging strand	Synthesised discontinuously, opposite to fork movement, in short Okazaki fragments. A primase lays down RNA primers, polymerase extends them, primers are removed and replaced with DNA, and DNA ligase joins the fragments.

To add a nucleotide, two of its three phosphates are removed; the energy released forms a **phosphodiester bond** attaching the remaining phosphate to the growing chain. DNA polymerases are extremely accurate - fewer than one error per 10^7 nucleotides - and many can **proofread**, removing mismatched bases from the strand's end.

3.4 DNA Repair and Mutations

Defects retained after proofreading are **mutations** - changes in the DNA base sequence arising from faulty replication or repair, or from physical and chemical agents. Agents that increase DNA damage or cell proliferation, raising the mutation rate, are **mutagens** (e.g. X-rays, gamma rays, UV light, acridine orange). Mutations may not be lethal but can disrupt regulation; a mutation in a somatic cell may cause uncontrolled division and cancer. Because any substance that raises the mutation rate also raises cancer risk, **all carcinogens are mutagens**.

General mechanism of repair

Template DNA carries methylated residues (N6-methyladenine, 5-methylcytosine) that the new strand lacks - letting enzymes recognise the correct strand.

Mismatched bases on the new strand are identified and removed along with a few neighbouring bases.

DNA polymerase beta synthesises a short correct segment, and DNA ligase seals the nick.

Diseases of defective DNA repair

Disorder	Key features
Xeroderma pigmentosum (XP)	Autosomal recessive; impaired nucleotide-excision repair of UV damage. Severe sunburn, freckling, pigmentation change, high skin-cancer risk (about half affected by age 10), cataracts and neurological problems. No cure - strict sun avoidance, sunscreen, protective clothing, retinoid creams and vitamin D. Life expectancy ~30 years below normal; affects ~1 in 100,000.
Ataxia-telangiectasia (A-T)	Autosomal recessive; caused by a defect in the ATM gene involved in recognising and repairing damaged DNA. Causes poor coordination (ataxia), dilated blood vessels (telangiectasia), immune weakness and raised cancer risk. Prevalence ~1 in 40,000 to 1 in 300,000.

3.5 Inhibitors of DNA Replication

DNA-synthesis inhibitors are antibiotics and drugs that target DNA synthesis. Many antibacterials - **novobiocin**, **nalidixic acid**, **ciprofloxacin** - inhibit the prokaryotic topoisomerase (DNA gyrase) and are widely used for urinary-tract and other infections. **Camptothecin**, an antitumour drug, inhibits human topoisomerase, and certain anticancer/antiviral agents block chain elongation by incorporating nucleotide analogues (e.g. 2',3'-dideoxyinosine).

3.6 The Central Dogma of Molecular Biology

The central dogma describes the flow of information in cells: information in DNA passes to messenger RNA, which directs the synthesis of a protein. It comprises three steps:

Step	What happens
Replication	Parent DNA is copied to form daughter DNA molecules with identical nucleotide sequences.
Transcription	The genetic message in DNA is rewritten as RNA; information is copied into mRNA in the language of nucleotides (A, C, G, U).
Translation	The mRNA message is decoded by ribosomes into a protein; the nucleotide sequence is translated into a sequence of the 20 standard amino acids.

3.7 Transcription

Transcription is the synthesis of single-stranded RNA from double-stranded DNA, transferring stored information into mRNA that directs protein synthesis. Its stages are **initiation**, **elongation**, **termination** and **post-transcriptional processing**. In mRNA, **Uracil replaces Thymine**, so the four nucleotides become A, G, C, U.

Template vs coding strand

Template strand 3'-CAGTTAGGC-5'

Coding strand 5'-GTCAATCCG-3'

mRNA transcript 5'-GUCAAUCCG-3'

The **template strand** is transcribed and is complementary to the mRNA. The **coding strand** has the same sequence as the mRNA (with T in place of U) and is complementary to the template.

3.8 Reverse Transcription

Although genes are normally made of DNA, the genetic material of some animal and plant viruses is RNA. **Retroviruses** - a subgroup of RNA viruses that includes HIV, the cause of AIDS - use their RNA as a template to synthesise DNA, catalysed by **reverse transcriptase** (an RNA-dependent DNA polymerase), producing double-stranded DNA from genomic RNA.

RT-PCR uses mRNA rather than DNA as the starting template: reverse transcriptase first makes a single-stranded complementary DNA (cDNA), DNA polymerase converts it to double-stranded DNA, and these molecules then serve as PCR templates. RT-PCR can determine whether an mRNA species is present, or clone a cDNA for later experiments.

3.9 Polymerase Chain Reaction (PCR) - In-vitro DNA Replication

PCR amplifies a chosen DNA sequence into millions of copies within a few hours. Invented by **Kary Mullis** (Nobel Prize 1993), it lets scientists take a tiny DNA sample and amplify it for detailed study, and is now indispensable in medical, clinical and forensic work. PCR relies on **thermal cycling** - repeated heating and cooling - using two main reagents: primers and a heat-stable DNA polymerase.

Step	Temperature & action
1. Denaturation	Strands are separated (melted) by heating to 95 degrees C for 15 s to 2 min.
2. Annealing	Cooling to ~50 degrees C lets primers hybridise with their complementary single strands.
3. Extension	Taq polymerase (from <i>Thermus aquaticus</i> , stable at high temperature) synthesises new strands at 72 degrees C for ~30 s in the presence of all four dNTPs.
4. Repeat	Cycles are repeated; the DNA doubles each cycle, so ~20 cycles give about a million-fold amplification, run automatically in a thermocycler.

Applications of PCR

Medical diagnostics (infectious disease, paternity), forensics, agriculture and ecology, biomedical research (gene expression, mutagenesis, methylation, sequencing), dentistry, archaeology and tissue typing.

3.10 Translation (Protein Biosynthesis) and the Genetic Code

DNA is transcribed to mRNA, which is translated into protein by ribosomes. The instructions are held in the mRNA as the **genetic code** - the system of nucleotide sequences that determines the amino-acid sequence of a protein. The code is read in triplets called **codons** (e.g. UUU codes for phenylalanine). Marshall Nirenberg received the 1968 Nobel Prize for deciphering it. There are 64 codons in total, including three **stop codons** (UGA, UAA, UAG).

Feature	Explanation
Triplet	Each codon is three consecutive bases (e.g. UUU = phenylalanine).
Non-overlapping	Codes are read consecutively; the starting point is critical.
Non-punctuated	There is no punctuation between codons - reading is continuous.
Degenerate	61 codons specify 20 amino acids, so most amino acids have more than one codon (serine has 6, glycine 4).
Unambiguous	Despite degeneracy, each codon specifies only one amino acid.
Universal	The same codon specifies the same amino acid in all species - 'the same for elephant and <i>E. coli</i> '.

Translation - the pathway of protein biosynthesis - converts the coded mRNA information into a new protein in five stages: **activation of amino acids**, **initiation**, **elongation**, **termination** and **post-translational processing**.

Inhibitors of protein synthesis

Inhibitor	Mechanism
Streptomycin	Binds the 30S subunit (prokaryotes) at the A site; blocks chain elongation by preventing aminoacyl-tRNA binding.
Tetracycline	Binds the 30S subunit; inhibits binding of aminoacyl-tRNA to mRNA (prokaryotes).
Chloramphenicol	Binds the 50S subunit; blocks the peptidyl-transferase reaction (prokaryotes).
Erythromycin	Binds the 50S subunit; inhibits translocation (prokaryotes).

Inhibitor	Mechanism
Lincomycin	Binds the 50S subunit; inhibits peptidyl transferase, preventing peptide-bond formation (prokaryotes).
Puromycin	Causes premature chain termination in both prokaryotes and eukaryotes.
Cycloheximide	Inhibits peptidyl-transferase activity of the 60S subunit (eukaryotes).

4 Recombinant DNA Technology and Cloning

4.1 Overview

Recombinant DNA technology (RDT) combines DNA from different sources to create new genetic combinations. Specific DNA segments are isolated and manipulated with enzymes, inserted into a host organism through a **vector** such as a plasmid, and then cloned to produce many copies of a desired protein. It is a fundamental tool of genetic engineering, used to produce therapeutic proteins, modify crops and diagnose genetic disease.

Genetic engineering is the direct manipulation of an organism's genes, including heritable and non-heritable recombinant DNA constructs. **Genetic recombination** is the exchange of information between two DNA segments; when a gene of one species is transferred into another organism by artificial means it is called recombinant DNA technology, commonly known as genetic engineering.

Some key definitions

Term	Definition
Gene expression	The process by which information in DNA becomes a functional product (protein or RNA); acts as an on/off switch and controls the 'volume' of production via transcription and translation.
Gene therapy	Treating or preventing disease by modifying, replacing or switching off faulty genes - often delivering functional genetic material with modified viral vectors.
Gene library	A collection of cloned DNA fragments from an organism, stored in bacterial or viral vectors - either genomic DNA (the whole genome) or cDNA (expressed genes).

4.2 Basic Principle and Fundamental Steps

Basic principle of RDT

Generate small DNA fragments of the genome and select the gene of interest.
Insert the gene into a vector (e.g. plasmid, cosmid) to create chimeric DNA.
Introduce the recombinant DNA into host cells.
Multiply and select the clone containing the recombinant molecule.
Express the selected clone to generate the desired product.

In practice the fundamental steps are: identify and isolate the desired gene; insert it into a suitable vector; introduce the vector into a host cell; allow the vector to multiply within the host, copying itself and the carried gene; pass copies to progeny as the host divides; and finally obtain a colony or **clone** of identical cells, each carrying the recombinant DNA molecule.

4.3 Key Tools for Cloning

Three tools are central: **restriction enzymes**, **DNA ligase** and **cloning vectors**, supported by polymerase enzymes, a host and a gene library.

Restriction enzymes (restriction endonucleases)

These bacterial enzymes cleave DNA at specific recognition sequences, producing discrete fragments with either cohesive (sticky) or blunt ends. They are part of the bacterial defence system against viruses, and are named after the genus and species of the organism from which they were isolated, with a number indicating the order of discovery. **EcoRI** was the first enzyme isolated from *Escherichia coli* strain RY13; **HindIII** was the third from *Haemophilus influenzae* strain Rd. For recombination, the same restriction site must be present on both vector and gene of interest.

Example - BamHI recognition site

BamHI recognises the double-stranded sequence:

5'--G G A T C C--3'

3'--C C T A G G--5'

It cuts between G and GATCC to leave complementary sticky ends.

Selected restriction enzymes and their sources

Microorganism	Enzyme	Recognition site (5' to 3')
<i>Bacillus amyloliquefaciens</i> H	BamHI	G [^] GATCC
<i>Brevibacterium albidum</i>	Ball	TGG [^] CCA
<i>Escherichia coli</i> RY13	EcoRI	G [^] AATTC
<i>Haemophilus aegyptius</i>	HaeII	PuGCGC [^] Py
<i>Haemophilus aegyptius</i>	HaeIII	GG [^] CC
<i>Haemophilus influenzae</i> Rd	HindII	GTPy [^] PuAC
<i>Haemophilus influenzae</i> Rd	HindIII	A [^] AGCTT
<i>Haemophilus parainfluenzae</i>	HpaI	GTT [^] AAC
<i>Haemophilus parainfluenzae</i>	HpaII	C [^] CGG
<i>Providencia stuartii</i> 164	PstI	CTGCA [^] G
<i>Streptomyces albus</i> G	Sall	G [^] TCGAC

DNA ligase

DNA ligase is the molecular glue that joins two DNA pieces from different sources through a covalent bond. Cells use ligation naturally during replication when Okazaki fragments are joined (up to a million breaks can occur in a single human cell each day). Ligase can join blunt ends, and can join sticky ends only when the overhanging nucleotides are complementary - which is why fragments to be joined are cut with the **same** restriction enzyme.

4.4 Cloning Vectors

Vectors are DNA molecules that carry a foreign DNA fragment to be cloned; they are self-replicating in an appropriate host. The most important are plasmids, bacteriophages, cosmids and artificial-chromosome vectors.

Vector	Characteristics
Plasmids	Extrachromosomal, double-stranded, circular, self-replicating DNA; ~0.5-5% of bacterial DNA. Carry resistance genes (e.g. ampicillin, tetracycline) that act as selection markers. Examples: pBR322, pUC.
Bacteriophages	Viruses that replicate within bacteria; some integrate into the bacterial chromosome. Carry larger fragments than plasmids - preferred for human genome work. Examples: phage lambda, phage M13.
Cosmids	Combine properties of both plasmid and phage lambda; carry larger foreign fragments than plasmids.
Artificial chromosomes	Accept very large fragments. Examples: human (HAC), yeast (YAC) and bacterial (BAC) artificial chromosomes.

Characteristic features of a cloning vector

An origin of replication (Ori) sequence

A selectable marker

One or more restriction sites

Antibiotic-resistance genes

Vectors and their insert sizes

Cloning vector	Insert size (kb)
Plasmid	0.5 - 8
Bacteriophage	5 - 25
Cosmid	35 - 45
BAC (bacterial artificial chromosome)	50 - 300
YAC (yeast artificial chromosome)	200 - 1000

4.5 Host Cells

Hosts are the living systems in which the vector is propagated; they may be prokaryotic or eukaryotic. Microorganisms are preferred because they multiply quickly. **E. coli** was the first organism used in DNA-technology experiments, but its drawback is that it cannot perform post-translational modifications. **Eukaryotic hosts** (e.g. mammalian cells) are preferred for human proteins because they can carry out the modifications needed for an active product - for example, tissue plasminogen activator.

4.6 General Steps of Cloning and Methods of Gene Transfer

General steps of cloning with any vector

Prepare the vector and target DNA by digestion with restriction enzymes to generate complementary ends.

Ligate the foreign DNA into the vector with DNA ligase.

Introduce the DNA into bacterial cells (or yeast for YACs) by transformation.

Select cells containing the foreign DNA by screening for a selectable marker.

Method of gene transfer	Description
Transformation	Uptake of plasmid DNA by E. coli in ice-cold CaCl ₂ (0-5 degrees C) followed by a heat shock (37-45 degrees C for ~90 s).
Conjugation	A natural microbial recombination process; plasmid-insert DNA is transferred from one cell to another.
Electroporation	High-voltage electric pulses induce cell membranes to fuse; liposome-mediated transfer (lipofection) uses lipid molecules with an aqueous interior to carry nucleic acids.
Direct transfer	DNA is delivered directly into the nucleus by microinjection or particle bombardment.

4.7 Applications of Recombinant DNA Technology

Area	Applications
Therapeutic proteins	Manufacture of interferon, plasminogen-activating factor, blood-clotting factors, insulin and growth hormone.
Diagnosis	Rapid AIDS testing; diagnosis of molecular diseases (sickle-cell anaemia, thalassaemia, familial hypercholesterolaemia, cystic fibrosis); prenatal diagnosis from amniotic fluid or chorionic villi.
Gene therapy	Cloning a gene into a vector taken up and incorporated into a host genome - ADA deficiency has been treated successfully.
Agriculture	Genetically engineered plants resistant to drought and disease, with improved food quality and increased crop yield.
Industry	Enzymes to produce sugars, cheese and detergents; protein products as food additives that raise nutritive value and flavour.
Forensic medicine	DNA fingerprinting - the restriction pattern is highly specific to each individual, helping identify criminals and settle parenthood disputes.

5 Biochemistry of Hormones

5.1 What is a Hormone?

A **hormone** is a chemical messenger secreted in trace amounts by one tissue and carried by the blood to a target tissue elsewhere, where it stimulates a specific biochemical or physiological activity. The word derives from a Greek verb meaning 'to excite'. Hormones are produced by special glands - adrenals, ovaries, parathyroids, pituitary, testes and thyroid - that secrete directly into the bloodstream and are therefore called **endocrine (ductless) glands**.

Characteristics of hormones

Strict specificity of biological action - replacement is impossible.

High biological activity - only a small quantity is needed.

Remote action - they act at a distance from the gland that made them.

5.2 Glands and the Roles of Hormones

Glands are cells or groups of cells that secrete substances. **Endocrine glands** spill hormones directly into the blood; **exocrine glands** route secretions (e.g. digestive enzymes, sweat) through ducts to specific destinations - for example, pancreatic amylase flows from the pancreas to the small intestine. Although hormone molecules pass billions of cells, they affect only specific **target cells** that carry matching receptor molecules, either embedded in the cell membrane or located in the cytoplasm. Binding activates a chain of events that expresses the hormone's effect.

Category	Role of hormones
Regulation	Control the internal environment by regulating secretion and excretion of chemicals such as salts and acids.
Response	Help the body respond to environmental change and cope with physical and psychological stress.
Reproduction	Control the reproductive cycle, conception, birth, sex-cell development and secondary sexual characteristics.
Growth & development	Essential to proper growth and development from conception to adulthood.

5.3 Classification of Hormones

Hormones are classified either by **chemical structure** or by **mechanism of action**.

By chemical structure

Class	Properties and examples
Peptide hormones	Water-soluble; 3 to over 200 amino-acid residues (e.g. hypothalamic and pituitary hormones, insulin, glucagon). Cannot cross the membrane - they bind surface receptors and raise production of a second messenger. Example: prolactin stimulating milk production; cyclic AMP often acts as the second messenger.

Class	Properties and examples
Steroid hormones	Fat-soluble (lipophilic) derivatives of cholesterol (e.g. adrenal cortical hormones, androgens, oestrogens). Pass freely through the membrane, bind cytoplasmic receptors, and the hormone-receptor complex enters the nucleus to trigger protein production.
Amine hormones	Small, water-soluble amino-acid derivatives containing amino groups (e.g. adrenaline of the adrenal medulla, thyroid hormones).

By mechanism of action (receptor location)

Group	Description
Group I	Lipophilic hormones (steroids; also thyroid hormones T3/T4) that cross the membrane and bind intracellular receptors - steroid receptors mainly in the cytoplasm, thyroid receptors in the nucleus.
Group II	Water-soluble peptide and amine hormones (except T3 and T4) that do not cross the membrane and bind cell-surface receptors, acting through second messengers.

5.4 Mechanism of Hormone Action

The first step is binding of the hormone to a specific receptor on the target cell. The hormone-receptor complex activates the receptor, which initiates the hormonal effect. Receptors are large, highly specific proteins, and this specificity is why a particular hormone acts only on a particular tissue.

Cytosolic / nuclear mechanism (Group I)

Lipid-soluble steroid and thyroid hormones cross the membrane and bind receptors in the cytoplasm or nucleus; the hormone-receptor complex is the intracellular messenger. The activated complex undergoes a conformational change and binds a regulatory DNA sequence called the **hormone response element (HRE)**, activating transcription of specific genes and the formation of mRNA. These mRNAs are translated into proteins that control the cell's metabolic functions.

Cell-membrane receptor mechanism (Group II)

Water-soluble hormones bind surface receptors and act through intermediary **second messengers** (the hormone itself being the first messenger). Binding causes a conformational change that activates a **G-protein** (a GTP-binding protein with alpha, beta and gamma subunits) by exchanging GDP for GTP on the alpha-subunit. In the absence of hormone the G-protein is inactive and bound to GDP.

The GTP-bound alpha-subunit dissociates and interacts with signalling enzymes such as **adenylate cyclase** (generating cAMP) or **phospholipase C** (generating inositol triphosphate, IP3, and diacylglycerol, DAG, from phosphatidylinositol). These second messengers mediate the hormone's action.

5.5 Second-Messenger Systems

cyclic AMP

cAMP, derived from ATP by adenylate cyclase, is the second messenger for many hormones (e.g. epinephrine, glucagon, calcitonin, PTH). It activates cAMP-dependent protein kinase,

which phosphorylates specific enzymes - activating some (e.g. glycogen phosphorylase) and inactivating others (e.g. glycogen synthase).

Phosphatidylinositol / calcium system

Certain hormone-receptor interactions activate phospholipase C through a specific G-protein. Phospholipase C breaks down membrane phosphatidylinositol bisphosphate (PIP₂) into two second messengers: **IP₃** and **DAG**. IP₃ liberates stored calcium from the mitochondria and endoplasmic reticulum; this calcium acts as a third messenger through calmodulin (a calcium-binding protein). DAG activates protein kinase C (PKC). Hormones such as TRH, gastrin and cholecystokinin act through this system.

Summary

Hormones are chemical messengers secreted into the blood to regulate other tissues.

Hormone action requires binding to a specific receptor.

Steroid and thyroid hormones use intracellular receptors and affect gene expression.

Water-soluble peptide and amine hormones (except T₃/T₄) use surface receptors and second messengers such as cAMP, Ca²⁺ and phosphatidylinositol.

6 Heme Degradation

6.1 Heme Structure, Function and Chemistry

Heme molecules are prosthetic coordination complexes produced in the liver and bone marrow that serve as the oxygen-binding component of several hemoproteins - a family of metalloproteins essential for tissue oxygenation. The heme complex in hemoglobin gives blood its red colour. Structurally, heme is a **porphyrin ring** bound as a tetradentate ligand to a central iron atom, with one or two axial ligands. Under normal conditions hemoglobin is intracellular, within red-cell cytoplasm; its release into plasma is typically a pathological event such as the hemolysis of sickle-cell disease.

The iron atom binds diatomic gases (including O₂) reversibly, causing conformational changes that favour the protein's biological function. Iron also serves as a source of electrons for heme-mediated redox and electron-transfer reactions. Binding selectivity depends on iron's oxidation state: the ferrous form Fe(II) complexes different ligands from the ferric form Fe(III). The conjugated porphyrin reduces tissue toxicity from singlet/radical electrons through delocalisation.

6.2 Tissue Deposition and the Need for Degradation

The hemoprotein family - defined by the presence of heme - includes endothelial nitric oxide synthase, catalases, cytochromes, heme peroxidase and myoglobin. Much heme is held in circulating erythrocytes complexed with hemoglobin, which releases **hemin** when ferrous Fe(II) is oxidised to ferric Fe(III). As red cells age or are damaged, splenic macrophages remove them, releasing further heme into plasma and tissues where it poses a toxicogenic, oxidative-stress challenge.

Because heme is lipophilic it can intercalate into lipid bilayers, physically and chemically disrupting organelles and cells, damaging the cytoskeleton, injuring endothelial cells and raising the risk of vascular inflammatory disorders. Left unchecked, these effects can contribute to chronic disease of the nervous system, heart, lungs, kidneys and liver.

6.3 Heme Oxygenase and the Degradation Pathway

The major adaptive countermeasure is degradation by **heme oxygenase (HO)**, a 32-kDa heat-shock protein (hsp32) of 288 amino-acid residues, found in the plasma membrane, nucleus and mitochondria, with especially high levels in the endoplasmic reticulum. Using NADPH, molecular oxygen and cytochrome P450 reductase, HO degrades heme to **biliverdin**, which biliverdin reductase converts to **bilirubin**, along with **carbon monoxide** and ferrous iron, Fe(II).

Two-step core reaction

Step 1 (heme oxygenase): $\text{Heme} \rightarrow \text{Biliverdin} + \text{Fe(II)} + \text{CO}$. The heme ring is opened at a specific carbon to give the linear tetrapyrrole biliverdin.

Step 2 (biliverdin reductase): $\text{Biliverdin (green)} \rightarrow \text{Bilirubin (red-orange)}$.

Fe(II) is quickly bound by ferritin; CO ensures about 1% of an individual's heme is complexed with CO even without environmental exposure.

6.4 Fate of Bilirubin

Bilirubin is largely insoluble and travels in blood bound to serum albumin. In the liver it is conjugated with glucuronate by bilirubin glucuronyl transferase to form **bilirubin diglucuronide**, which is water-soluble enough to be secreted in bile into the small intestine. There, microbial enzymes convert it mainly to **urobilinogen**. Some urobilinogen is reabsorbed and carried to the kidney, where it becomes **urobilin**, giving urine its yellow colour; the remainder is converted to **stercobilin**, which gives faeces its red-brown colour.

Product	Significance
Biliverdin	Green tetrapyrrolic bile pigment; intermediate converted to bilirubin.
Bilirubin	The most abundant antioxidant in mammalian tissue and a major cytoprotective agent, especially important in the developing newborn brain. Excess (beyond albumin's solubilising capacity) is toxic.
Carbon monoxide	Toxic at high concentration, but at the low levels of heme degradation acts as a vasodilator (less potent than nitric oxide) with effects on neurotransmission.
Ferrous iron, Fe(II)	Toxicogenic; sequestered by ferritin (stored as Fe(III)) and transported by transferrins to sites of use such as red-cell progenitors.

6.5 Jaundice and Clinical Correlates

Impaired liver function or blocked bile secretion causes bilirubin to leak into the blood, yellowing the skin and eyes - a condition called **jaundice**. The cell toxicity of jaundice may reflect bilirubin levels exceeding the serum albumin available to solubilise it. Measuring blood bilirubin helps diagnose underlying liver disease.

Neonatal jaundice

Newborns may develop jaundice because they have not yet produced enough glucuronyl bilirubin transferase to process bilirubin.

A traditional treatment - exposure to a fluorescent (phototherapy) lamp - photochemically converts bilirubin into more soluble, easily excreted compounds.

6.6 Regulation of Heme Degradation

Given the varied roles of its products, the degradative pathway is regulated mainly at the first step. Humans have at least three heme-oxygenase isozymes. **HO-1** is highly regulated and induced by a wide range of stresses (shear stress, angiogenesis, hypoxia, hyperoxia, heat shock, UV light, hydrogen peroxide and other metabolic insults). **HO-2** is expressed continuously, mainly in brain and testes. **HO-3** is not yet well characterised.

Summary of heme degradation

Degradation begins in splenic macrophages, which remove senescent red cells; globin is recycled to amino acids and heme is processed.

Heme oxygenase opens the ring to give biliverdin, Fe(III) and CO; biliverdin reductase then forms bilirubin (green → red-orange).

Bilirubin is carried by albumin to the liver, conjugated with glucuronate and excreted in bile.

In the intestine it is deconjugated to urobilinogen and stercobilin; some urobilinogen is reabsorbed and excreted as urobilin in urine, while most is oxidised to stercobilin in faeces.

7 Xenobiotic Metabolism (Part 1)

7.1 Introduction to Xenobiotics

A **xenobiotic** is a chemical substance foreign to animal life - including plant constituents, drugs, pesticides, cosmetics, flavourings, fragrances, food additives, and industrial and environmental pollutants. It is estimated that humans are exposed to 1-3 million xenobiotics in their lifetimes. Most enter via diet, air, drinking water, drug administration and lifestyle choices, and undergo detoxification processes that generally render them less toxic, more polar and readily excretable. Foreign molecules are also produced in the intestine by bacterial enzymes acting on digestion products.

Xenobiotics can produce pharmacological responses, toxicity, immunological reactions and cancer. Some are harmless but can still provoke biological responses, which often depend on the conversion of the absorbed substance into an active metabolite.

7.2 Detoxification and Biotransformation

Most foreign compounds are metabolised in the body. The processes leading to their disposition were once called **detoxification**, but the term is not always appropriate because the products are sometimes more toxic than the original substance - so **biotransformation** is preferred. The primary purpose of biotransformation is to convert a lipophilic, non-polar, toxic compound into a more polar, water-soluble form, reducing its permeability through the lipid cell membrane and facilitating excretion in urine or bile.

When detoxification turns harmful

Vinyl chloride is biotransformed to vinyl chloride epoxide, which binds covalently to DNA and RNA - a step toward liver cancer.

Acetaminophen is normally biotransformed and excreted safely, but at high doses the enzymes saturate and the excess follows a pathway producing a liver-toxic metabolite. Acetaminophen toxicity is treated with N-acetylcysteine.

In general, lipophilic xenobiotics are converted to more polar, readily excreted products - lipophilic barbiturates such as thiopental and pentobarbital would have extremely long half-lives without this conversion.

7.3 Sites of Drug Metabolism

Although every tissue can metabolise, the **liver** is the principal organ of drug metabolism; the GIT, lungs, skin and kidneys also show considerable activity. After oral administration many drugs (e.g. isoproterenol, meperidine, pentazocine, morphine) are absorbed from the small intestine and carried via the portal system to the liver for extensive metabolism - the **first-pass effect**. Some drugs (clonazepam, chlorpromazine, cyclosporine) are metabolised more in the intestine than the liver, while midazolam undergoes significant (~50%) intestinal metabolism.

The first-pass effect can so limit availability (e.g. lidocaine) that alternative routes must be used. Drugs may also be metabolised by gastric acid (e.g. penicillin), by digestive enzymes (e.g. insulin), by enzymes in the intestinal wall, or by intestinal micro-organisms. Most

transformations are catalysed by enzymes in the endoplasmic reticulum, mitochondria, cytosol, lysosomes, nuclear envelope or plasma membrane.

7.4 Microsomal Mixed-Function Oxidase and Cytochrome P450

Many enzymes lie in the lipophilic ER membranes of the liver. When these membranes are isolated by homogenisation and fractionation they re-form into vesicles called **microsomes**. Rough microsomes are dedicated to protein synthesis; smooth microsomes are rich in oxidative drug-metabolising enzymes - the **mixed-function oxidases (MFO)** or mono-oxygenases, which require a reducing agent (NADPH) and molecular oxygen. Two microsomal enzymes are central:

Key components

A flavoprotein, NADPH-cytochrome P450 reductase - one mole contains one mole each of FMN and FAD.

A hemoprotein, cytochrome P450, which serves as the terminal oxidase.

The overall mono-oxygenase reaction can be written:



Cytochrome P450 enzymes are heme-containing mono-oxygenases in the liver ER that absorb light at 450 nm when exposed to carbon monoxide - hence the name. **NADPH** (not NADH) is the coenzyme; electrons pass from NADPH to P450, reductively activating molecular oxygen so that one oxygen atom is inserted into the substrate. There are about 150 P450 isoforms, classified by structural homology into family, subfamily and individual enzyme, and they are inducible.

Nomenclature example - CYP2D6

CYP = Cytochrome P450

2 = Family number

D = Subfamily letter

6 = Individual isoform / gene number

Major human hepatic P450s

Isoform	Approx. share of liver P450
CYP1A2	~15%
CYP2D6	~4%
CYP2C9	~20%
CYP2E1	~50%
CYP3A4	~30%

Together these catalyse the bulk of hepatic xenobiotic metabolism. Notably, **CYP3A4 alone** metabolises over 50% of clinically prescribed drugs handled by the liver. Individuals vary in metabolic rate, partly through differences in CYP expression in liver and intestine, influenced by diet, tobacco and alcohol.

7.5 Phases of Detoxification

Most transformations occur between absorption into the general circulation and renal elimination, and fall into two major categories: **Phase I** and **Phase II** reactions.

Phase	Nature of the reaction
Phase I	Alters the foreign molecule by adding a functional group (-OH, -NH ₂ , -SH), usually converting the parent compound to a more polar metabolite. Often decreases toxicity, but can increase it (e.g. methanol to formic acid). Products may be excreted directly or proceed to Phase II.
Phase II	Conjugation (synthetic) reactions in which an endogenous substrate - glucuronic acid, sulphuric acid, acetic acid or amino acids - combines with the functional group to form a highly polar, easily excreted conjugate.

Although Phase I usually precedes Phase II, the order can reverse: the hydrazide moiety of **isoniazid** forms an N-acetyl conjugate (Phase II), which is then hydrolysed to isonicotinic acid (Phase I). A subset of products may also undergo **Phase III** reactions.

7.6 Phase I Reactions

Phase I reactions include oxidation, reduction and hydrolysis.

A. Oxidation

Alcohols, aldehydes, amines, aromatic hydrocarbons and certain drugs are destroyed by oxidation. **Hydroxylation** is the chief oxidative reaction, catalysed by mono-oxygenases / cytochrome P450:



Oxidation type	Examples
Alcohols / aldehydes to acids	Ethanol → acetic acid; methanol → formic acid; benzaldehyde → benzoic acid; benzyl alcohol → benzoic acid.
Aromatic hydrocarbons to phenols	Benzene → phenol; acetanilide → aminophenol (later conjugated).
Oxidative deamination of amines	Butylamine → acetoacetic acid; benzylamine → benzoic acid (nitrogen to ammonia).
Oxidative dealkylation	Toluene → benzoic acid.

B. Reduction

Less common than oxidation. Aldehydes may be reduced to alcohols; for example, picric acid → picramic acid.

C. Hydrolysis

Esters, amides and glycosides are hydrolysed by esterases, amidases and glycosidases: procaine → p-aminobenzoic acid + diethylaminoethanol; aspirin → salicylic acid + acetic acid; atropine → tropic acid + tropine. Other Phase I reactions include epoxidation, O- and N-dealkylation, N- and S-oxidation, deamination, desulphuration, dechlorination, and azo-, nitro- and carbonyl reductions.

7.7 Phase II Reactions

In Phase II, parent drugs or their Phase I metabolites bearing suitable chemical groups undergo **conjugation** with an endogenous substrate to yield a drug conjugate. Conjugation makes the compound more polar and water-soluble, and therefore more easily excreted in urine or bile. Many conjugating agents are known: glucuronic acid, glycine, glutamine, cysteine, glutathione, acetyl-CoA, PAPS (3'-phosphoadenosine-5'-phosphosulfate) and SAM (S-adenosylmethionine).

Types of Phase II conjugation

Conjugation	Endogenous reactant	Transferase / location	Substrates & example drugs
Glucuronidation	UDP-glucuronic acid	UDP-glucuronosyl transferase / microsome	Phenols, alcohols, sulphonamides, hydroxylamines; morphine, chloramphenicol, salicylic acid, indomethacin, dapsone.
Acetylation	Acetyl-CoA	N-acetyltransferase / cytosol	Amines; sulfonamides, isoniazid.
Glutathione conjugation	Glutathione (GSH)	GSH-S-transferase / cytosol or microsomes	Epoxides, nitro groups, hydroxylamines; acetaminophen.

Glutathione conjugation

Conjugation with glutathione (GSH) is the commonest Phase II reaction.

Glutathione (gamma-glutamyl-cysteinyl-glycine) is a tripeptide of glutamic acid, cysteine and glycine.

A wide range of xenobiotics are excreted following GSH conjugation.

8 Xenobiotic Metabolism (Part 2)

8.1 Phase II Reactions (Continued)

Further Phase II substrates include aromatic nitro compounds, halogenated compounds and aliphatic halides. The general glutathione reaction is:



where X is the xenobiotic, GSH is glutathione and X-S-G is the glutathione conjugate. Glutathione conjugates are further metabolised and excreted as **mercapturic acids** (sulfur-containing compounds).

Additional conjugation pathways

Conjugation	Active form	Transferase / location	Substrates & examples
Glycine conjugation	Glycine	Acyl-CoA glycine transferase / mitochondria	Salicylic acid, benzoic acid, nicotinic acid, folic acid.
Sulfation	PAPS (phospho adenosyl phosphosulfate)	Sulfotransferase / cytosol	Phenols, alcohols, aromatic amines; estrone, phenol, aniline, 3-hydroxycoumarin, acetaminophen.
Methylation	SAM (S-adenosylmethionine)	Methyltransferase / cytosol	Phenols, alcohols, aromatic amines, catecholamines; dopamine, epinephrine, histamine, pyridine, thiouracil.
Water conjugation	Water	Epoxide hydrolase / microsome, cytosol	Arene oxides, mono- and cis-disubstituted oxiranes; benzopyrene-7,8-epoxide, leukotriene A ₄ .

8.2 Some Important Phase I and Phase II Reactions

Worked sequences

Example 1: Toluene $\rightarrow$ (NADPH) benzyl alcohol $\rightarrow$ benzoic acid.

Example 2 (both phases): Benzene $\rightarrow$ (Phase I) phenol $\rightarrow$ (Phase II) phenyl sulfate.

Oxidation of alcohol groups

Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) $\rightarrow$ (alcohol dehydrogenase, $\text{NAD}^+ \rightarrow \text{NADH} + \text{H}^+$) acetaldehyde (CH_3CHO); then acetaldehyde $\rightarrow$ (aldehyde dehydrogenase) acetic acid (CH_3COOH). Similarly, benzaldehyde $\rightarrow$ (aldehyde dehydrogenase) benzoic acid.

Reductive reactions and hydrolysis

Phase I reduction: nitrobenzene $\rightarrow$ (NADPH $\rightarrow$ NADP $^+$) aminobenzene (aniline). Acetylation: aniline + acetic acid $\rightarrow$ acetanilide + H_2O . Hydrolysis: aspirin (acetylsalicylic acid) + H_2O $\rightarrow$ salicylic acid + acetic acid.

Acetylation (Phase II / Phase III)

Isoniazid $\rightarrow$ (N-acetyltransferase + acetyl-CoA) acetylated isoniazid; sulfanilamide $\rightarrow$ (N-acetyltransferase + acetyl-CoA) acetyl-sulfanilamide.

8.3 General Pathway of Drug Metabolism

Mapping the route

Phase I: Drug → oxidised drug; oxidised drug → reactive metabolites → toxicity.

Phase II: Oxidised drug → conjugation (detoxification) → excretion from the body.

Alternatively, drugs may enter detoxification directly via Phase II conjugation, leading to excretion.

8.4 Phase III Reactions

Phase III reactions are uncommon; a typical example is further conjugation with glutathione. Most xenobiotics entering the body are drugs, detoxified by the drug-metabolising enzymes. Induction of the cytochrome P450 system can produce unwanted effects in some people - for example, induction of delta-aminolevulinic acid (ALA) synthase by barbiturates precipitates attacks in acute intermittent porphyria. In some cases P450-dependent oxygenases convert xenobiotics into harmful compounds (e.g. benzopyrene becomes a carcinogen by epoxidation). Drug-metabolising enzymes also show genetic variation - decreased, increased or absent activity - so slow metabolisers may show toxic effects at normal doses.

8.5 Factors Affecting Drug Metabolism

The duration and intensity of a drug's action depend largely on the rate at which it is metabolised to inactive products. Several factors modify the activity of drug-metabolising enzymes.

Factor	Effect on metabolism
Enzyme induction	Increased synthesis (or decreased degradation) of enzymes in response to a drug, mainly by inducing transcription of cytochrome P450 mRNA. May yield a less active metabolite (decreased activity) or a more toxic one (increased activity).
Enzyme inhibition	Inhibitors bind the active site, reducing enzyme-substrate complex formation and product. More important clinically than induction, especially for drugs with a narrow therapeutic index. Examples: allopurinol, aspirin, erythromycin, ketoconazole.
Environmental chemicals	Halogenated pesticides, cigarette smoke, insecticides and heavy metals. Charcoal-broiled foods and cruciferous vegetables induce CYP1A; grapefruit juice inhibits CYP3A4. Smokers metabolise drugs faster through enzyme induction.
Age and sex	Microsomal enzymes are underdeveloped in neonates and the elderly. Caffeine has a half-life of 4 days in neonates versus 4 hours in adults; the elderly have smaller livers, lower enzyme activity and reduced hepatic blood flow, so need lower doses.
Diet	A high-protein diet increases metabolism (raising enzymes and amino acids for conjugation); a low-protein diet decreases it. Grapefruit juice inhibits metabolism and improves oral bioavailability; starvation reduces glucuronide formation.
Sex differences	Often due to sex hormones. Male rats have greater drug-metabolising capacity (androgens); women metabolise benzodiazepines more slowly than men, and women on contraceptive pills metabolise several drugs more slowly.